



Application Note:

LR2021

FLRC Improvements

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1 Introduction

Fast Long Range Communication (FLRC) is one of the technologies available in the LR2021 and provides a similar link budget advantage to LoRa® but at higher data rates.

LoRa is a spread spectrum modulation technique, which represents a single bit of information as multiple chips within a LoRa chirp. It provides the user with the ability to trade-off data rate for receiver sensitivity (which equates to range). To increase the LoRa data rate the designer must either reduce the number of chips representing each bit, which is achieved by lowering the spreading factor or by increasing the LoRa bandwidth.

LoRa works well at low data rates in the world's license-free industrial, scientific, and medical (ISM) bands however, due to the scarcity of spectrum, such bands are typically only a few MHz wide. Because it is a spread spectrum modulation, with low spectral efficiency, employing LoRa at higher data rates implies using significant transmission bandwidths.

The use of spread spectrum techniques is less appropriate at higher data rates in the ISM bands, so other technologies are used to extend the radio's communication range beyond that possible with legacy modulation schemes – FLRC addresses this scenario. This note presents the basics of FLRC, its configuration, and expected performance in practical scenarios.

2 FLRC Benefits

FLRC is a modulation format based upon coherent Minimum-Shift Keying (MSK) and provides the following benefits:

- High data rates: FLRC provides the same 5 dB link budget improvement (versus FSK) as LoRa for higher data rates. FLRC delivers data rates up to 2.6 Mbps versus LoRa which can be used up to 125 kbps.
- High spectral efficiency: FLRC reduces spectral occupancy and does not use spread spectrum compared with LoRa such that the raw bit rate equals the bandwidth, before the application of Forward Error Correction (FEC).
- Low energy performance: The FLRC transmitter uses the same high efficiency PA as the LoRa modem and can operate at receiver currents as low as 6.6 mA at 260 kbps.

The following graph portrays the performance of FLRC relative to the Shannon limit for a fixed noise figure, and the same performance advantage experienced with LoRa is continued into higher data rates by FLRC.

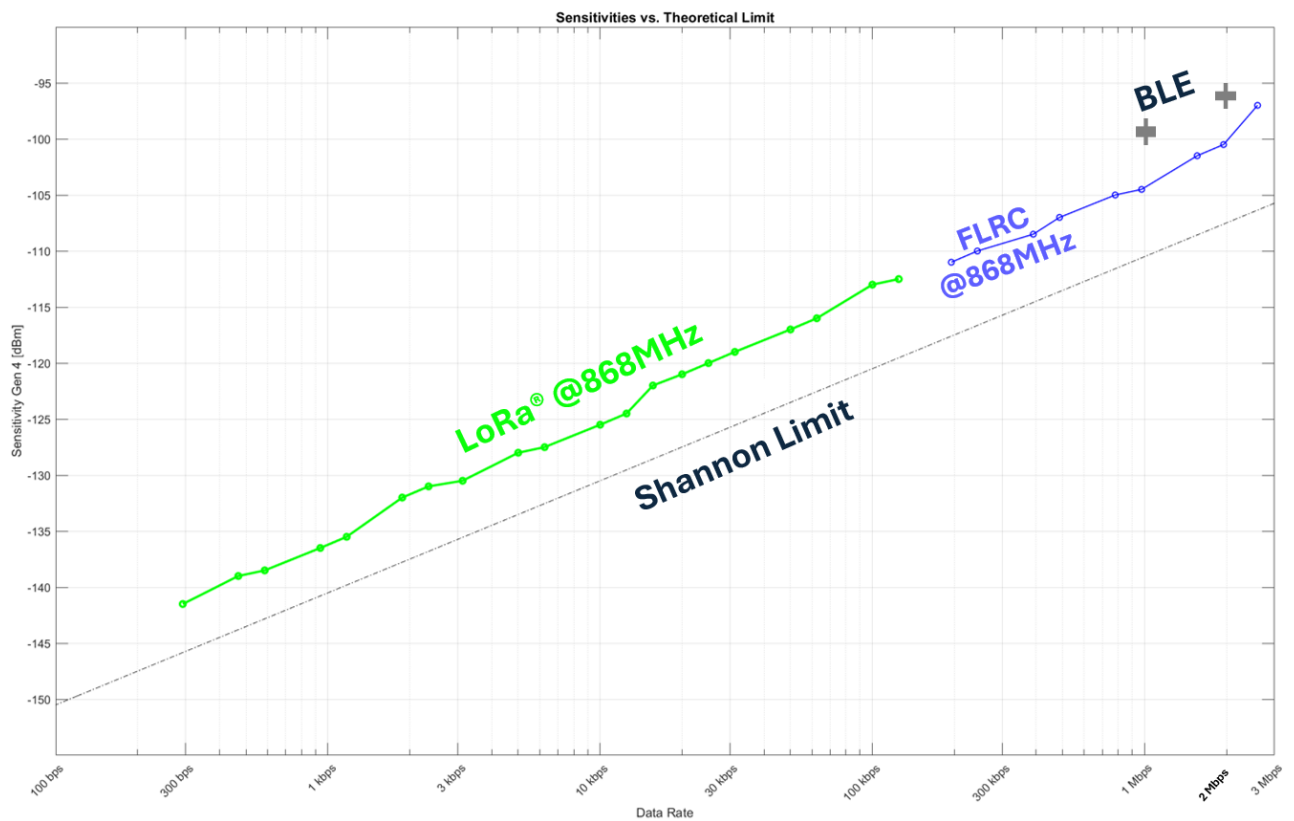


Figure 1. The Performance of FLRC Relative to LoRa, BLE and the Shannon Limit.

3 Key Applications

FLRC is suited to applications where long range, high spectral efficiency, high energy efficiency and low latency are required such as the following:



High density sensor networks: FLRC permits high data rates which provides lower time-on-air. Because the same traffic can be transmitted in less time this gives higher network capacity.



Edge AI Sensors: Many edge sensing devices employ artificially intelligent classifiers to identify an event that is occurring based upon raw sensor input data. This allows the use of a long-range, low data rate communications such as LoRa to send the event information without having to send *all* the raw data. However, new or unknown events may produce data that is difficult to classify. In such cases, a high data rate channel allows the transmission of raw sensor data to allow human operator intervention, or refinement of the classifier model by providing additional training data.



Real time remote control and audio: The high data rates and throughputs that FLRC affords are high enough to permit the transmission of audio data over longer ranges than possible with conventional modulation schemes at the same data rates. A high data rate also helps to reduce latency making it ideal for remote control applications or applications that require real time, immediate feedback to the user



Image Transfer: By splitting (or fragmenting) an image across multiple packets, it becomes possible to transfer static images from battery powered end devices, over long ranges, using the high data rates accessible with FLRC.

4 Core Technologies

The performance enhancement of FLRC compared to FSK is achieved using two main techniques:

- **Coherent GMSK demodulation:** Coherent GMSK demodulation provides a 5 dB sensitivity improvement over conventional FSK.
- **Convolution coding:** Employs Forward Error Correction (FEC) and redundancy within the packet to improve sensitivity at the limit budget and in environments where interference is present

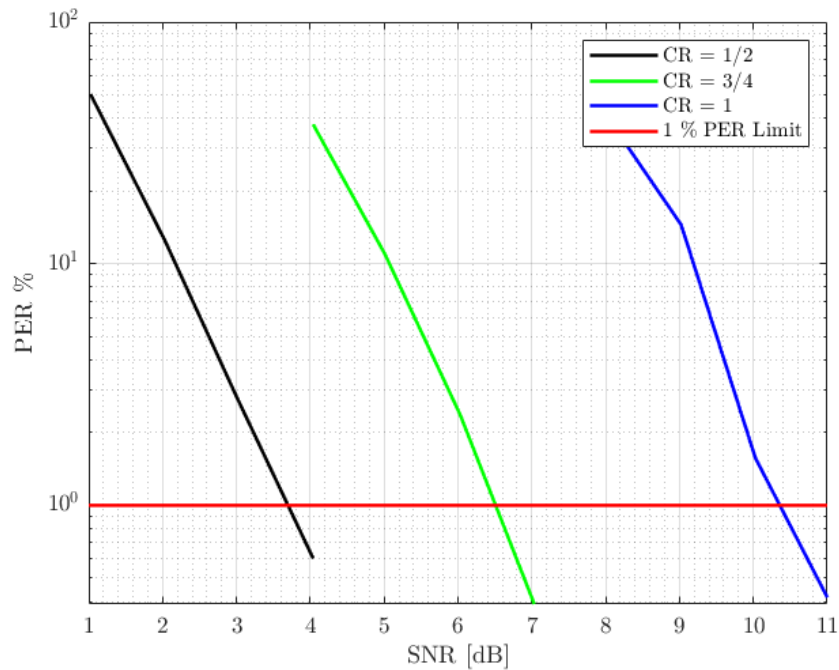


Figure 2. Influence on coding rate on sensitivity

The coding rate is an important parameter for the designer to consider when designing an FLRC link as it has an influence on sensitivity and the probability that the receiver will wake-on noise. This is discussed in more detail in Section 5.3.

5 FLRC Best Practices

To give the designer control over the link performance, FLRC provides a wide range of modem settings that allow the radio to be tuned for the best compromise between range, energy efficiency and data throughput. This Section describes those settings.

5.1 FLRC Sensitivity

The FLRC modem is available in both low and high frequency bands of operation of the LR2021. The following tables highlight a subset of the range of modem settings available to the user, which includes the influence of the effective data rate and time on air for a 31 byte packet. The high frequency front-end is typically used in the 2.4 GHz ISM band and the lower frequency used in the sub-GHz ISM bands (868 MHz in Europe, 915 MHz in the US and 490 MHz in China). For compliance information see [1].

Table 1. Sensitivity specification high frequency (>1 GHz).

Bit Rate [Mbps]	Bandwidth [MHz]	CR	Sensitivity [dBm]	Effective Bit Rate [Mbps]	ToA (31 Byte) [us]
2.6	2.6	$\frac{3}{4}$	-99	1.95	151
1.3	1.2	$\frac{3}{4}$	-101.5	0.957	231
0.65	0.6	$\frac{3}{4}$	-104	0.488	463
0.26	0.3	$\frac{3}{4}$	-108.5	0.195	1157

Table 2. Sensitivity specification lower frequency (<1 GHz).

Bit Rate [Mbps]	Bandwidth [MHz]	CR	Sensitivity [dBm]	Effective Bit Rate [Mbps]	ToA (31 Byte) [us]
2.6	2.6	$\frac{3}{4}$	-100.5	1.95	151
1.3	1.2	$\frac{3}{4}$	-104.5	0.957	231
0.65	0.6	$\frac{3}{4}$	-107	0.488	463
0.26	0.3	$\frac{3}{4}$	-111	0.195	1157

5.2 Frequency Error Tolerance

The FLRC modem automatically assigns the receive bandwidth necessary to accommodate the transmitted FLRC signal. However, it is also necessary to consider the frequency offset between the transmitter and receiver caused by reference oscillator frequency errors.

The following table shows the permissible offset as a function of both data rate and payload length for 10% packet error rate. The dependency on payload length arises because during the transmission process the crystal is subject to instantaneous heating. However, this effect can be avoided by keeping the total frequency error between Tx and Rx less than the frequency limits below.

Table 3. Measured Typical Frequency Error Tolerance [kHz] of the FLRC Modem versus Data Rate for all Frequency Bands.

Payload Length (Bytes)	2600 kbps [kHz]	2080 kbps [kHz]	1300 kbps [kHz]	1040 kbps [kHz]	650 kbps [kHz]	520 kbps [kHz]	325 kbps [kHz]	260 kbps [kHz]
10	±206	±235	±142	±121	±76	±59	±37	±30
31	±206	±226	±137	±117	±73	±57	±36	±29
100	±206	±212	±128	±111	±69	±53	±34	±27
200	±206	±205	±122	±107	±67	±51	±32	±26
300	±206	±202	±118	±104	±66	±50	±32	±25
400	±205	±201	±117	±104	±65	±50	±32	±25
511	±206	±198	±119	±102	±64	±50	±31	±25

The full range of tolerances is presented here to give the designer the freedom to potentially relax the crystal specification in applications where the maximum payload size will not be used.

To ensure that the frequency tolerance for each band is explicitly clear, here the same quantities are listed in PPM for the 868/915 MHz and 2.4 GHz ISM bands.

Table 4. 868/915 MHz Band PPM Typical Frequency Error Tolerance of the FLRC Modem versus Data Rate.

Payload Length (Bytes)	2600 kbps [PPM]	2080 kbps [PPM]	1300 kbps [PPM]	1040 kbps [PPM]	650 kbps [PPM]	520 kbps [PPM]	325 kbps [PPM]	260 kbps [PPM]
10	±164	±164	±109	±109	±77	±55	±33	±27
31	±225	±257	±155	±132	±83	±65	±41	±32
100	±225	±247	±150	±128	±80	±63	±40	±31
200	±225	±231	±140	±121	±75	±58	±37	±29
300	±225	±224	±133	±117	±73	±56	±35	±28
400	±225	±221	±129	±114	±72	±55	±35	±28
511	±224	±219	±128	±113	±71	±54	±34	±28

Table 5. 2.45 GHz Band PPM Typical Frequency Error Tolerance of the FLRC Modem versus Data Rate.

Payload Length (Bytes)	2600 kbps [PPM]	2080 kbps [PPM]	1300 kbps [PPM]	1040 kbps [PPM]	650 kbps [PPM]	520 kbps [PPM]	325 kbps [PPM]	260 kbps [PPM]
10	±84	±96	±58	±49	±31	±24	±15	±12
31	±84	±92	±56	±48	±30	±23	±15	±12
100	±84	±86	±52	±45	±28	±22	±14	±11
200	±84	±84	±50	±44	±27	±21	±13	±10
300	±84	±82	±48	±42	±27	±21	±13	±10
400	±84	±82	±48	±42	±26	±20	±13	±10
511	±84	±81	±48	±42	±26	±20	±13	±10

5.3 FLRC Wake-on-Noise

Because FLRC is designed for the reception of much higher data rates than LoRa and the throughput of data with much less preamble and header overhead - the probability of false detection of packets on random thermal noise is increased. It can be prudent to reduce the probability of false detection of packets on noise, especially in applications where this could affect system performance. In this Section we describe which settings can influence this.

5.3.1 Packet Parameter Selection

One important consideration is how the FLRC packet should be configured. The two packet formats available in FLRC mode, fixed length (no header) and variable length (with header) are illustrated below.

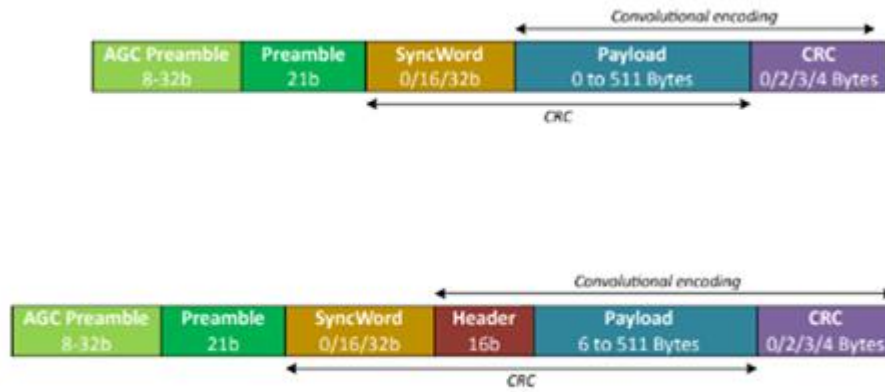


Figure 3. The FLRC Packet Format.

Although the fixed or variable mode selection has no influence on detection probability: the packet features an 8- to 32-bit Automatic Gain Control (AGC) preamble followed by a fixed 21-bit preamble, a payload of up to 511 bytes, and cyclical redundancy check (CRC). The length of each of these fields has an influence on improving the immunity to false packet detection.

To reduce the probability of false detection on noise, each of these fields should be set to the maximum value, i.e.:

- **32-bit Preamble**
- **32-bit Sync Word**
- **4 Byte CRC**

5.3.2 Convolutional Coding Rate Selection

Whilst the packet fields have an intuitive effect on reducing the likelihood of false detection on noise: longer fields equating to better protection. The FEC coding Rate, CR, can also impact this performance.

The FEC adds redundant bits to the transmitted packet so the receiver can use this information to assist in recovering corrupted data. This results in an increase in the observed sensitivity as the redundancy in the packet increases. Figure 3 shows that convolutional coding applies only to the header, payload, and CRC portions of the packet. To realise this increase in sensitivity, the correlation detection threshold must be placed at a lower level than for lower redundancy coding rates.

The lower detection threshold required for the higher sensitivity detection, increases the probability of reception on noise as shown below:

Table 6. FEC Influence on Receiver Detection on Noise

CR	1	3/4	2/3	1/2
Redundancy	None	33%	50%	100%
P(false detect)	Low	Low	Med	High

6 FLRC Range Performance

Range testing is one method of demonstrating optimum performance relative to other modulation schemes. The following results compare the performance of FLRC in several scenarios.

6.1 FLRC vs LoRa

A free space range test provides a reliable indication of the link budget at a given frequency and an indication of the performance in unobstructed environments. The test in Figure 4 shows the range of FLRC at 2.4 GHz 260 kbps, versus LoRa at 2.4 GHz 476 bps (SF12 200 kHz) and 868 MHz at 292 bps (SF12, 125 kHz). (Note, both transmitter and receiver were located at 1.5 m, so the LoRa range is terrain limited).

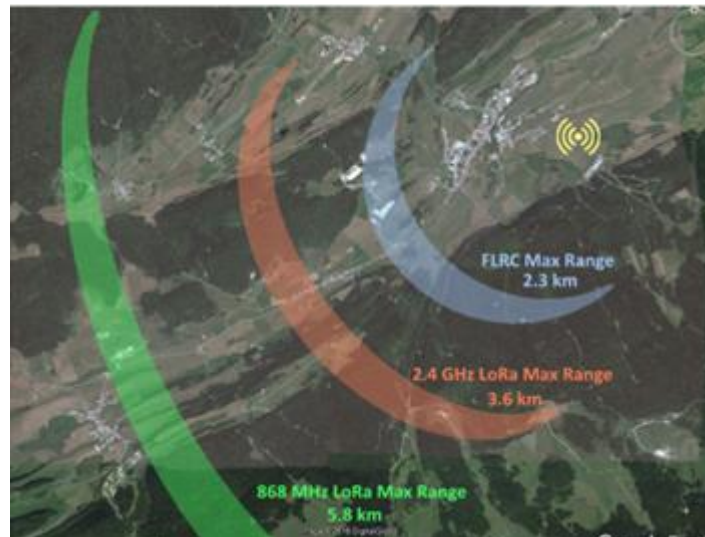


Figure 4. 2.4 GHz FLRC vs 2.4 GHz and sub-GHz LoRa.

6.2 Sub-GHz FLRC Range

One of the new features available in LR2021 is the ability to communicate using FLRC in the sub-GHz band. The reduced pathloss experienced in the sub-GHz bands permits high data rate communication at ranges well beyond equivalent data rates in the 2.4 GHz band.

In the following figures we see tests performed using 2.6 Mbps and 260 kbps data rates at 868 MHz. In the first result we see reliable communication up to 1.3 km, beyond which the radios are non-Line of Sight (nLoS) so communication is lost.

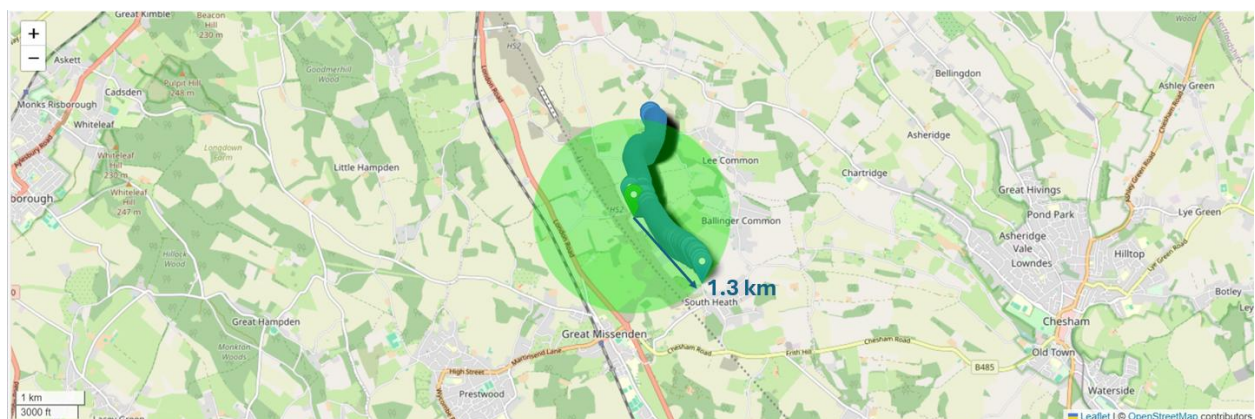


Figure 5. Range Test at 868 MHz 2.6 Mbps CR = 1/2.

The same test at 260 kbps shows a large range increase in the nLoS region, up to over twice the range possible at 2.6 Mbps.

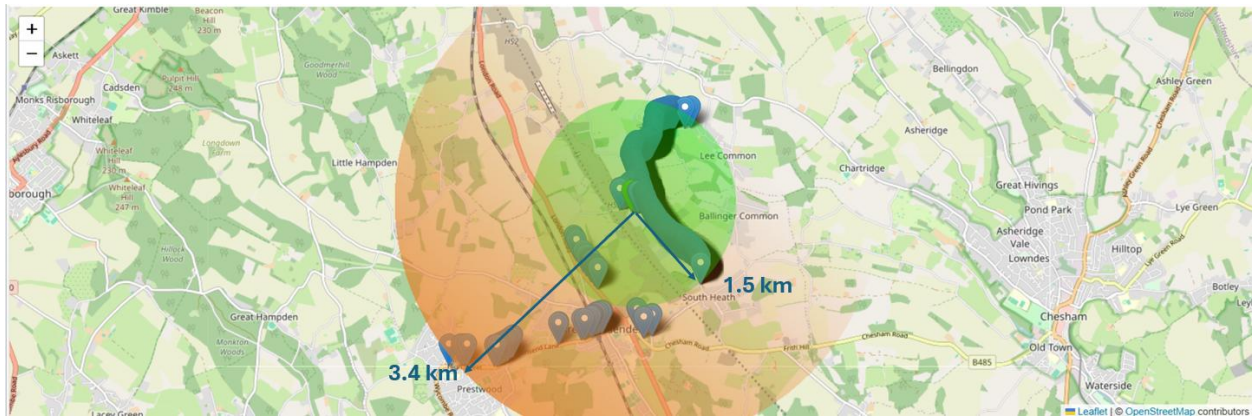


Figure 6. Range Test at 868 MHz 260 kbps CR = 2/3.

6.3 Indoor Range Test

FLRC also shows significant range improvement in comparison to Bluetooth low energy (BLE). Two Indoor tests are featured that indicate the performance of two identical packet transmissions made with a BLE data rate of 1 Mbps.

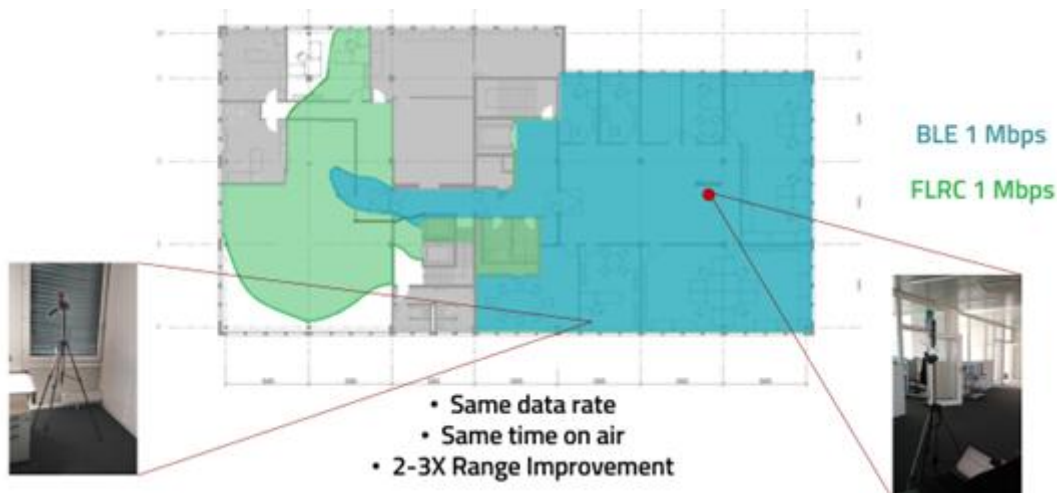


Figure 7. 2.4 GHz Indoor Range Test

FLRC can extend the range beyond the conventional BLE user experience for exactly the same data rate, operating at distances of twice that possible with conventional BLE.

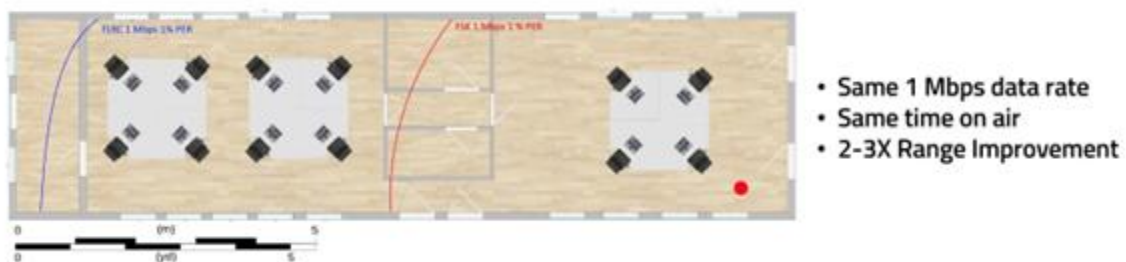


Figure 8. Another Indoor Range Test versus BLE.

7 Conclusion

FLRC extends the LoRa performance advantage of 5 dB additional link budget performance, relative to FSK, to higher data rates. For low energy consumption applications, the energy saving is doubled which means:

- Minimal time spent in transmission due to high data rates.
- Low consumption when in FLRC reception and transmission modes.

Approaches to increase the immunity to waking the receiver on noise were presented, including the selection of coding rates CR1 and CR3/4, in addition to maximising the protections afforded by the FLRC packet fields.

Range testing reveals that FLRC provides a significant range increase compared to using FSK at equivalent data rates, for example, Bluetooth range is doubled when substituting FLRC.

The FLRC modem is available for the first time ever in the sub-GHz band in LR2021, whilst retaining full compatibility with SX1280 devices. It provides the user a high data rate communication physical layer (PHY) with sensitivity improvement previously only possible with LoRa.

8 References

[1] AN1200.112 "FCC and ETSI Certification of the LR2021"

[2] Semtech Corporation, "LR2021 Long Range, Low Power, Multi-band LoRa® Transceiver Datasheet," Rev. 1.1, Semtech Corporation, Camarillo, CA, USA, Oct. 2025. [Online]. Available: <https://www.semtech.com/products/wireless-rf/lora-plus/lr2021>

9 Revision History

Version	ECO	Date	Changes and/or Modifications
1.0	ECO-076642	July 2026	Initial Version



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